



P4: Hotel Reservation System

Database Management System Mini Project — a fully functional relational database for managing hotel operations including room booking, customer management, and billing.

MICROSOFT SQL SERVER

T-SQL

SSMS

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Problem Statement & Objectives

Design a robust, normalized database system to manage end-to-end hotel operations — from guest check-in to billing — with full data integrity and automation.

Scalable Schema

Normalized up to BCNF for clean, efficient data organization

Data Integrity

Enforced via PRIMARY KEY, FOREIGN KEY, UNIQUE, and CHECK constraints

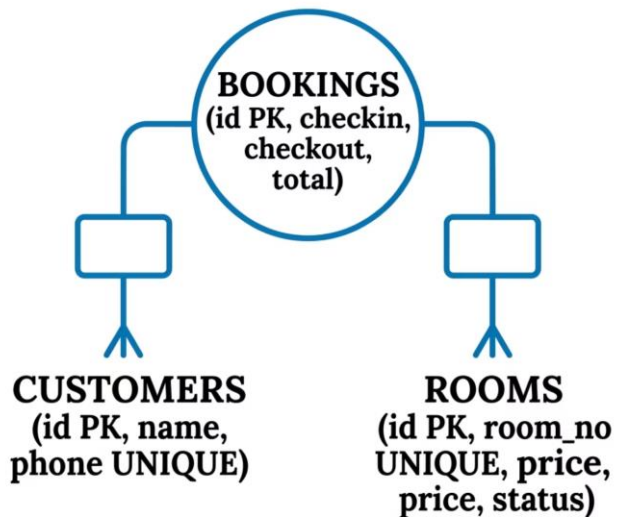
Automation

Triggers handle total calculation and room status updates automatically

Security & Control

Transaction control (TCL) and role-based access (DCL) fully implemented

Database Schema & Relationships



BCNF Achieved. The schema ensures data integrity and eliminates redundancy.

Three Core Tables

- **Customers** — id (PK), name, phone (UNIQUE)
- **Rooms** — id (PK), room_no (UNIQUE), price, status
- **Bookings** — id (PK), cust_id (FK), room_id (FK), checkin, checkout, total

Relationships

- Customers → Bookings: **One-to-Many**
- Rooms → Bookings: **One-to-Many**

③ Normalization achieved up to **BCNF (Boyce-Codd Normal Form)**, eliminating all redundancy and update anomalies.

Feature Overview

Core SQL Features

DDL

CREATE, DROP, ALTER

Constraints

PK, FK, UNIQUE,
NOT NULL, CHECK

DML

INSERT, UPDATE, DELETE

Indexes

Optimized query performance

Advanced SQL Features

Views

vw_AvailableRooms, vw_AllBookings

Triggers

trg_CalcTotal, trg_UpdateRoom

TCL & DCL

COMMIT, ROLLBACK, GRANT, REVOKE

Functions

fn_Nights — calculates stay duration

Joins & Subqueries

INNER JOIN, LEFT JOIN, GROUP BY,
HAVING

7

SQL Script Files

3

Tables

30+

Total Operations

2

Triggers

Query Demonstrations

1

INNER_JOIN — Customer Room Mapping

Joins Customers → Bookings → Rooms to display guest names with assigned room numbers.

Output: Raj $\rightarrow$ Room 101 | Priya $\rightarrow$ Room 201

2

VIEW — Available Rooms Report

Queries `vw_AvailableRooms` to list all unbooked rooms with pricing.

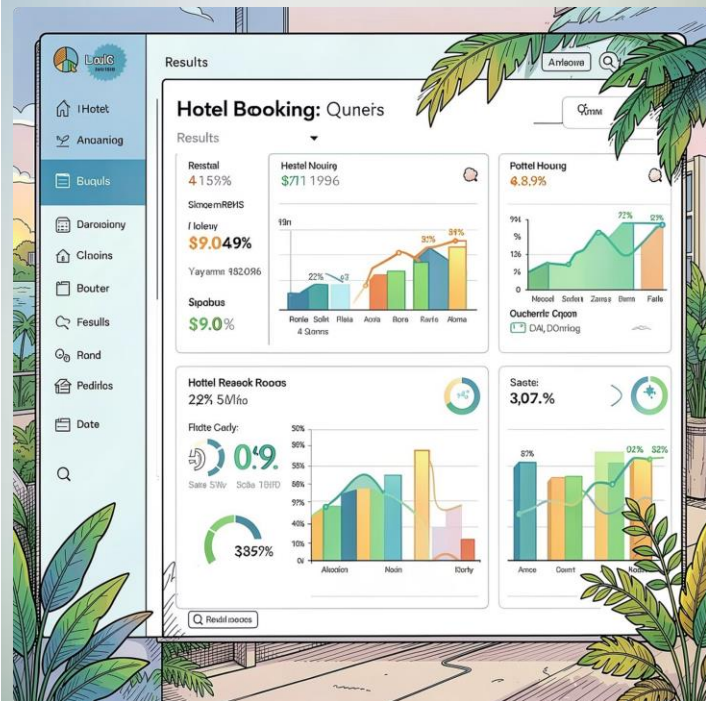
Output: Rooms 101, 102, 201, 202, 301 with prices ₹3000–₹8000

3

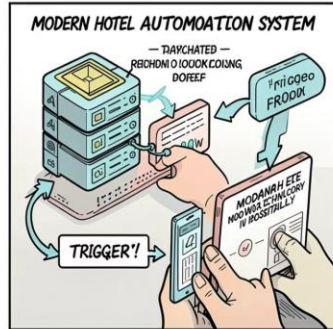
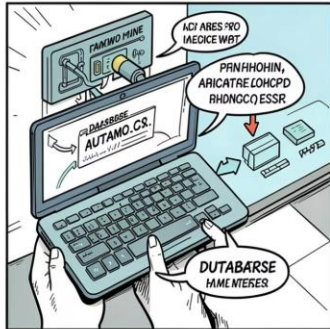
SUBQUERY — Booked Customers

Uses a nested **SELECT** to retrieve names of all customers with active bookings.

Output: Raj, Priya, Amit



Functions & Triggers



fn_Nights — Scalar Function

Inputs: check-in date, check-out date • **Returns:** number of nights

Example: `fn_Nights('2026-06-01', '2026-06-03')` → **2 nights**

trg_CalcTotal — BEFORE INSERT Trigger

Fires on Bookings insert. Auto-calculates total = Nights × Room Price, eliminating manual entry errors.

trg_UpdateRoom — AFTER INSERT Trigger

Fires after booking confirmation. Updates room status from 'Available' to 'Booked', preventing double bookings.



Automation Flow: Booking Insert → Calculate Total → Update Room Status → Complete

Transaction Control & Data Security

Transaction Control (TCL)

Ensures data consistency through atomic operations — either all changes commit or none do.

Data Control (DCL)

Role-based access control restricts user permissions to protect sensitive data.

```
CREATE USER hotel_user WITHOUT LOGIN;  
GRANT SELECT ON vw_AvailableRooms TO hotel_user;  
REVOKE INSERT ON Customers FROM hotel_user;
```

 **Result:** hotel_user has read-only access to room reports — no insert or delete permissions on core tables.

COMMIT

```
BEGIN TRANSACTION; INSERT INTO Bookings...; COMMIT;
```

Data permanently saved to database.

ROLLBACK

```
BEGIN TRANSACTION; DELETE FROM Bookings...; ROLLBACK;
```

Deletion cancelled, data fully restored.

Summary & Future Scope

✔ What We Built



Database normalized to **BCNF** with 7 tested SQL scripts



Full **DDL**, **DML**, **DCL**, **TCL** implementation



Joins, Subqueries, and Views operational



2 automated Triggers + 1 scalar Function



Index created for query performance

🔮 Future Enhancements

💳 Payment Gateway

Online payment integration for seamless checkout

📊 Live Dashboard

Real-time room availability and occupancy analytics

📱 Mobile App

Customer-facing app for booking and check-in

🏆 Loyalty Program

Reward points and repeat-guest benefits

Project Repository: github.com/PrashastiSrivastava/